

Fungal Infections

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Introduction :

- Some fungi live on the skin as part of the normal skin flora while others come into contact with the skin through the environment and animals.
- Superficial fungal infections attack the epidermis, mucosa, nails, and hair, and are divided into two groups: **1) Molds** (e.g. dermatophytes) and **2) Yeasts** (e.g. Candida).
- **Yeast** that comprises part of the normal skin flora can become pathogenic because of a change in the host's immune system >> this allows the yeast to disseminate throughout the body >> causing serious life-threatening diseases.
- Conversely, **mycoses** that originate as systemic infections can become deposited in the skin via hematogenous spread.

Investigating fungal infections :

- Taking skin scrapings, nail clippings, scalp brushings, and skin biopsies for mycological analyses.
- Moistened bacterial swabs taken from the affected skin and inoculated into standard fungal media can be a useful additional test.
- Macroscopic patterns of fungal growth from **cultures** may also lead to species diagnoses; however, **fungi are usually slow to grow.**
- Modern fungal diagnostics are moving towards the use of rapid **PCR** tests and **ELISA**.

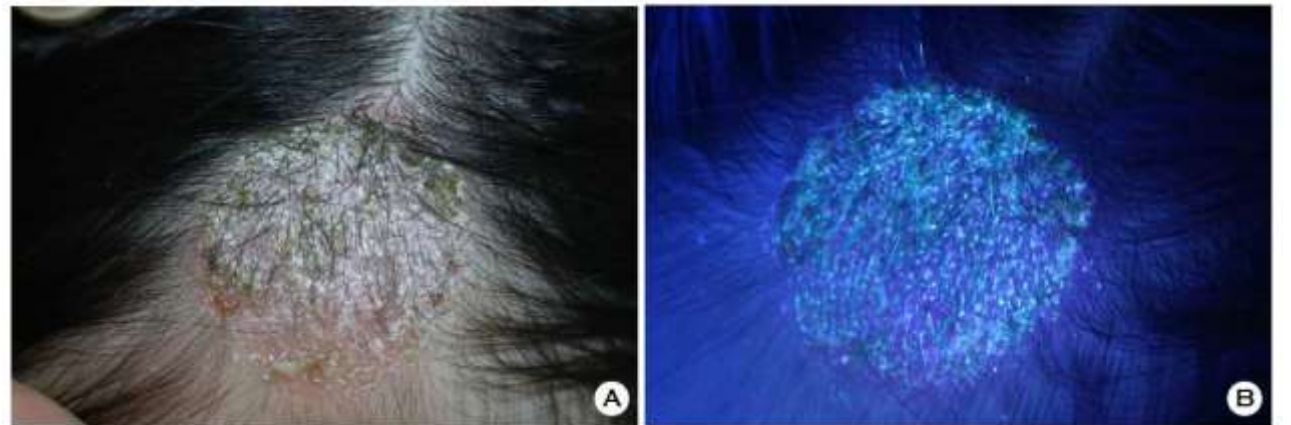
Principles of diagnosis:

- Consider a fungal infection in any patient with **itchy, dry, scaly** lesions (usual distribution is asymmetrical).
- Skin samples are taken by **scraping the edge** of the lesion with a blunt blade held at right angles to the skin.
- Nail clippings should be taken from the nail including subungal debris.



Principles of diagnosis:

- Laboratories will report initially on direct microscopy but **culture results take 2–4 weeks.**
- Lesions to which steroids have been applied are often quite clinically atypical because the normal inflammatory response is suppressed – **tinea incognito.**
- Wood's light (ultraviolet light) can reveal Microsporum infections of hair, as they produce **a green-blue fluorescence.**



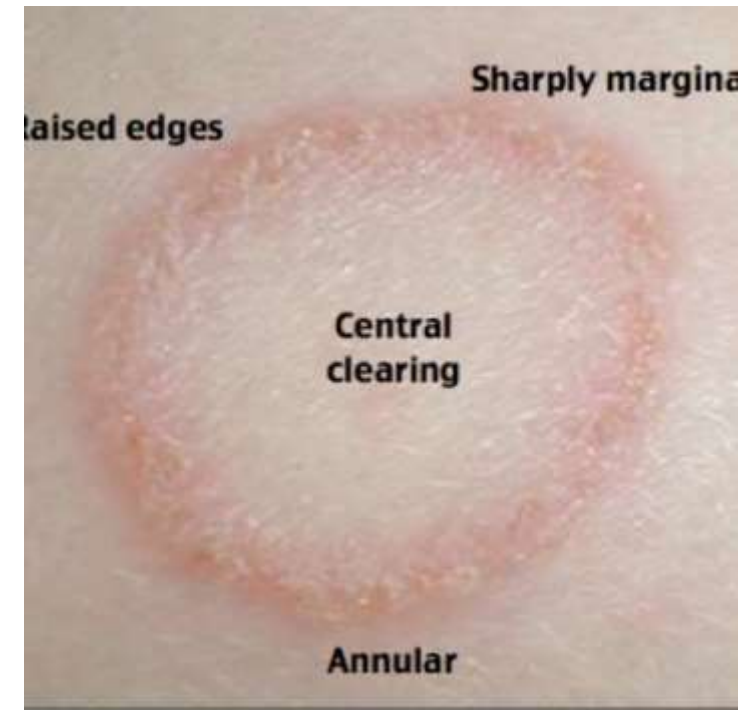
General features of fungi in the skin:

- Superficial dermatophyte infections are named according to the body site affected:
 - Tinea capitis (scalp)
 - Tinea corporis (body)
 - Tinea cruris (groin)
 - Tinea pedis (feet).
- Fungal infections usually cause **itching**; the skin may be **dry** and **scaly** or in flexural areas wet maceration can result.
- Zoophilic (animal) fungi generally produce a more intense inflammatory response with deeper indurated lesions than fungal infections due to anthropophilic (human) species.



General features of fungi in the skin:

- Some lesions have a prominent scaling margin with apparent clearing in the center, leading to annular or ring-shaped lesions – hence the term ‘ringworm’.
- **Tinea capitis:** usually affects **Children** below the age of puberty (due to low sebum which contains antimicrobials).
- **Tinea pedis:** typically affects **Adults**.
- **Tinea cruris:** seen mainly in **men**
- **Onychomycosis:** is particularly common in the **elderly** and **debilitated**.



Scalp and Face:

Tinea capitis (scalp ringworm):

- Mainly affects pre-adolescent children.
- The main fungal pathogens isolated include Trichophyton, Microsporum, and Epidermophyton.
- The most commonly isolated fungus in urban settings is **T. tonsurans**; it penetrates the hair shaft (endothrix fungus) and is characterized by single or multiple patches of alopecia, often minimal scaling and occasionally inflammation.
- T. tonsurans must be treated with **systemic antifungal** therapy to clear the endothrix infection, 1st line is **Griseofulvin**.



Tinea capitis:

- Clinically, features are highly variable: diffuse scaling, grey patches, black dots (broken-off hairs), multiple pustules, patchy alopecia, extensive alopecia with inflammation, kerion formation, and occipital lymphadenopathy.
- **A kerion** is an inflamed, boggy, pustular lesion on the scalp that occurs when there is a brisk inflammatory response. This settles with systemic antifungal treatment and does not require surgical drainage.





Figure 16.5 Kerion in *T. tonsurans* tinea capitis.



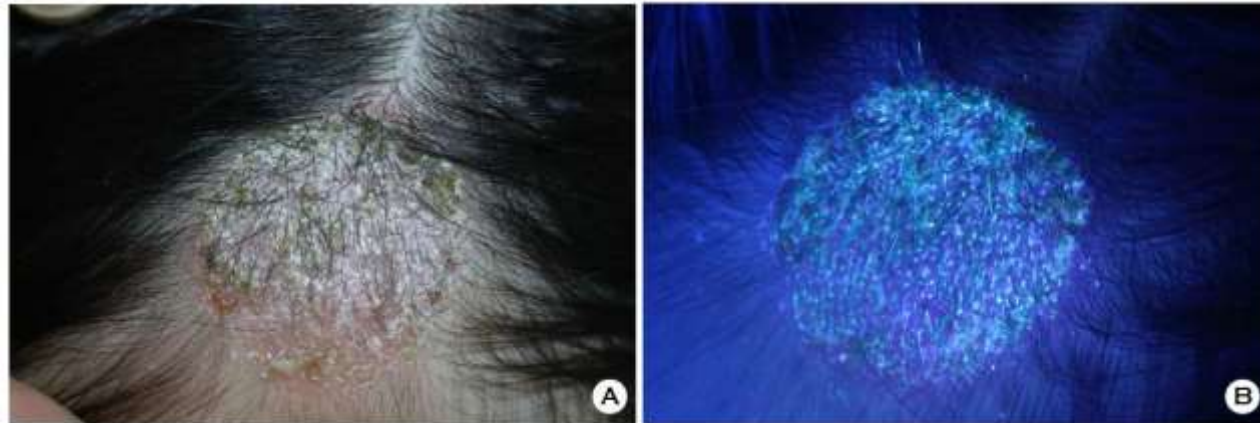
Figure 16.4 Extensive alopecia and inflammation caused by *T. tonsurans* infection.

Tinea Capitis Differential Diagnosis :

- Scalp eczema/psoriasis
- Folliculitis
- Alopecia areata
- Seborrheic dermatitis

Tinea Capitis Diagnosis :

1. Wood's light (ultraviolet light) infections of hair, as they produce **a green-blue fluorescence**.
2. Scalp Brushing >>> mycology analysis



Tinea Capitis Treatment :

- Topical antifungals are unable to penetrate the hair shaft sufficiently to eradicate endothrix infections and therefore systemic antifungal agents are required for rapid clinical and mycological cure.
- **Oral griseofulvin** (10mg/kg if over 1 month of age; occasionally 20mg/kg is required) is the current **FDA-approved treatment**.
- Occasionally, a papular/pustular widespread cutaneous eruption appears after the commencement of systemic antifungal treatment – this is a so-called '**id reaction**' which is an immunological response and not an adverse drug reaction.



Tinea incognito:

- A term used for the indistinct appearance of a superficial fungal infection on the skin caused by the **use of topical/systemic steroids**.
- Therefore, because **the typical clinical features of the fungal infection** (raised scaly margin with inflammation) **are lost**, the diagnosis becomes more difficult.
- Management is to stop the steroids and treat them with topical antifungal agents.



Figure 16.7 Tinea incognito.



Feet and Hands:

Tinea pedis or athlete's foot:

- A common disease mainly affecting adults.
- It is easily acquired in public swimming pools or showers, and industrial workers appear to be particularly predisposed to this infection.
- **Very itchy** and can affect any part of the foot but frequently occurs between the toes (especially the **fourth toe web**) where the skin becomes macerated.
- Across the plantar and dorsal aspects of the feet there is usually a dry, scaling rash occasionally with vesicles at the active margins.



Tinea pedis or athlete's foot:

- The hands may be similarly affected.
- The condition needs to be differentiated from psoriasis and eczema (particularly pompholyx) and therefore scrapings for mycology can be helpful.
- Treatment: Terbinafine 1% cream twice daily for 2–4 weeks is usually effective but recurrent infections may occur.



Trunk: (Tinea corporis, Tinea cruris)

Tinea corporis:

- Also causes pruritus.
- Lesions tend to be erythematous with a **well-defined** scaly edge.



Tinea cruris:

- In the groin
- Usually symmetrical dry scaling and may spread to the upper inner thighs.
- DDx:
 - 1- Candidal yeast infection (intense erythema and satellite lesions)
 - 2- Corynebacterium minutissimum (Coral red fluorescence in Wood's Light, require a systemic erythromycin/tetracycline)
 - 3- Psoriasis
 - 4- Mycosis fungoides
 - 5- Eczema.



Tinea cruris

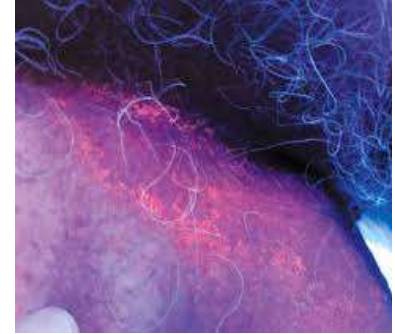


Candidal yeast infection

Tinea cruris:

- DDx:
 - 1- Candidal yeast infection (intense erythema and satellite lesions)
 - 2- *Corynebacterium minutissimum*, erythrasma (Coral red fluorescence in Wood's Light, require a systemic erythromycin/tetracycline)
 - 3- Psoriasis
 - 4- Eczema.

erythrasma



Eczema



Inverse Psoriasis

Treatment tinea cruris or Tinea corporis:

- Terbinafine 1% cream is the most effective topical treatment for tinea corporis/cruris
- Other agents include miconazole, clotrimazole, ketoconazole and econazole
- Duration of treatment: 2–4 weeks.
- If systemic therapy is required, itraconazole (100mg daily) or terbinafine (250mg daily) for 2 weeks is usually effective.

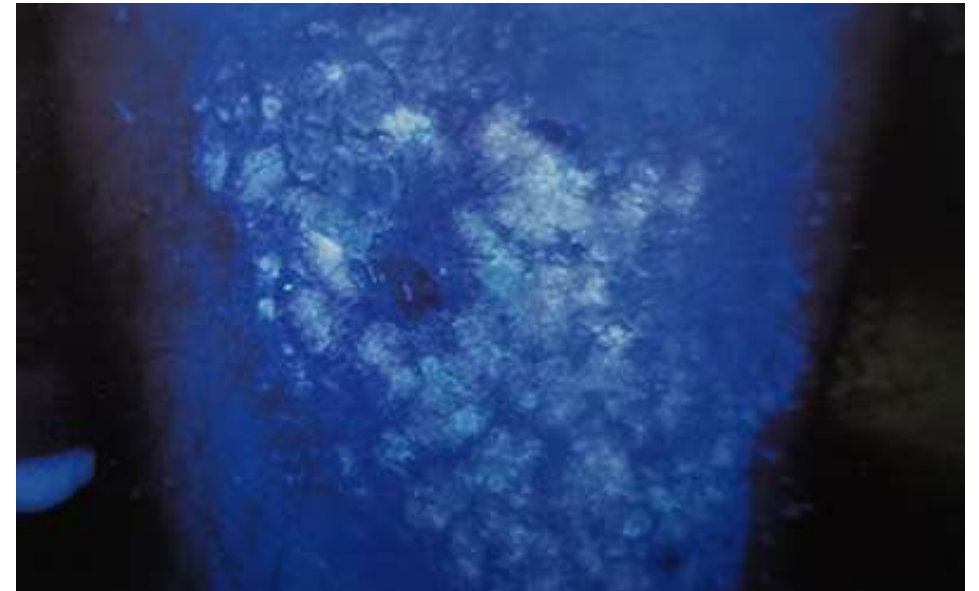
Pityriasis versicolor:

- Affects the upper back, neck, chest, and arms.
- It usually becomes apparent when the skin is exposed to the sun as these areas fail to tan.
- Well-defined macular lesions of variable color (hence the name ‘versicolor’) from darker brown to pale tan colored fine scale appear.



Pityriasis versicolor:

- Diagnosis by:
 - 1- Stretch test
 - 2- Wood's Light exam: Golden yellow color
 - 3- In skin scrapings, the causative organism **M. furfur** can be readily identified.
- Treatment: **Topical** selenium sulfide and topical ketoconazole 2% cream applied once daily for 2 weeks reportedly cures between 70 and 80% of patients, but one-third relapse.
- The pale areas left behind after treatment will only re-pigment once the patient is re-exposed to sunlight.



Pityriasis versicolor:

Differential diagnosis:

- Seborrheic dermatitis
- Pityriasis rosea
- Guttate psoriasis
- Vitiligo.



Seborrheic dermatitis



Pityriasis rosea



Guttate psoriasis

Vitiligo



Nails Onychomycosis :

- Affects mainly **adult toenails** and is usually caused by **dermatophytes**.
- Nail plates become thickened, brittle, and white to yellow/brown.
- The distal nail plate is usually affected initially with spread proximally to involve the nail fold.
- Onychomycosis may be caused by **yeast** infections such as *Candida albicans*.



Nails Onychomycosis ddx:

- **Psoriasis of the nail**: the changes occur proximally and tend to be symmetrical and are associated with pitting and other evidence of psoriasis elsewhere.
- **Lichen planus**: may also cause nail dystrophy but this is usually manifested by vertical ridging and nicks in the nails.





Figure 16.16 *Candida* onychomycosis.



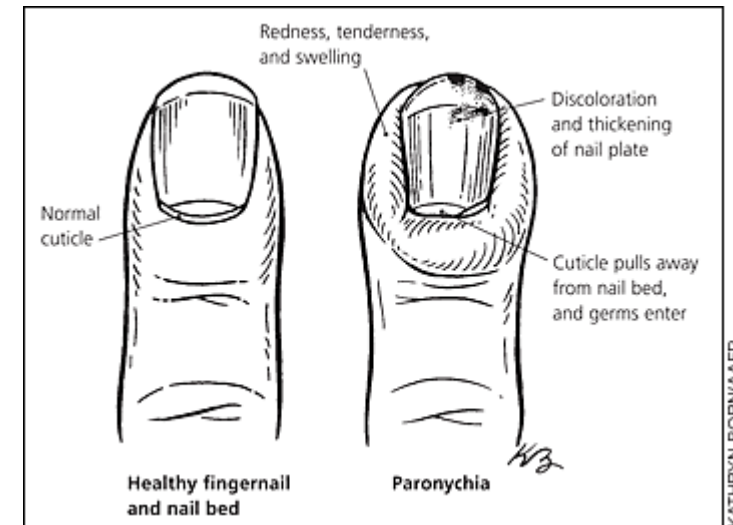
Figure 16.15 Onychomycosis caused by *Trichophyton rubrum*.

Nails Onychomycosis Treatment:

- **Systemic therapy** with terbinafine 250mg daily for 16 weeks (toenails) or 8 weeks (fingernails) is usually considered the **first line**.
- **Topical treatment:** should be considered for a single nail or very mild distal nail-plate onychomycosis. Agents available include amorolfine and ciclopirox olamine 8% nail lacquer solutions, sodium pyrithione, bifonazole/urea, imidazoles, and allylamines.
- Diseased nail plates **may be ‘dissolved’** using 40% urea preparations applied carefully to the nail plate, occluded with clingfilm left on overnight, and then scrapped off before repeating each night for about 2 weeks. This helps to physically remove the infected nail without the need for medications or surgery. As the nail plates regrow, a topical antifungal such as amorolfine should be used twice weekly to prevent re-infection.

Chronic paronychia:

- Occurs around the nails of individuals involved in ‘wet-work’ who repeatedly put their hands in water (such as child carers, chefs, doctors, dentists, nurses, and hairdressers).
- **Other RF:**
 - DM, poor peripheral circulation, removal of the cuticle, and artificial nails.
- There is erythema and swelling of the nail fold, loss of cuticle, with brownish discoloration of the nail.
- **Treatment:** The hands should be kept as dry as possible, and an azole lotion applied regularly around the nail fold, Oral antifungal can be used.



Yeast infections :

Candida Intertrigo:

- Candida infection may occur in the **flexures** of infants, elderly or immobilized patients, especially under the breasts and abdominal skin folds.
- This should be differentiated from (a) psoriasis, which does not usually itch; (b) seborrhoeic dermatitis, a common cause of a flexural rash in infants; and (c) contact dermatitis/discoid eczema.
- Candida intertrigo is symmetrical and ‘satellite’ pustules or papules outside the outer rim of the rash are typical.



Figure 16.17 Candida infection in the groin.

Yeast infections:

- Yeast, including *C. albicans*, may be found in the mouth and vagina of healthy individuals.
- Clinical lesions in the mouth – white **buccal plaques or erythema** – may develop.
- **Predisposing factors** include impaired immunity (including HIV), diabetes mellitus, endocrine disorders, and corticosteroid treatment.
- **Vaginal candidosis** or **thrush** is a common (occasionally recurrent) infection of healthy young women, leading to itching, soreness, and a mild discharge.

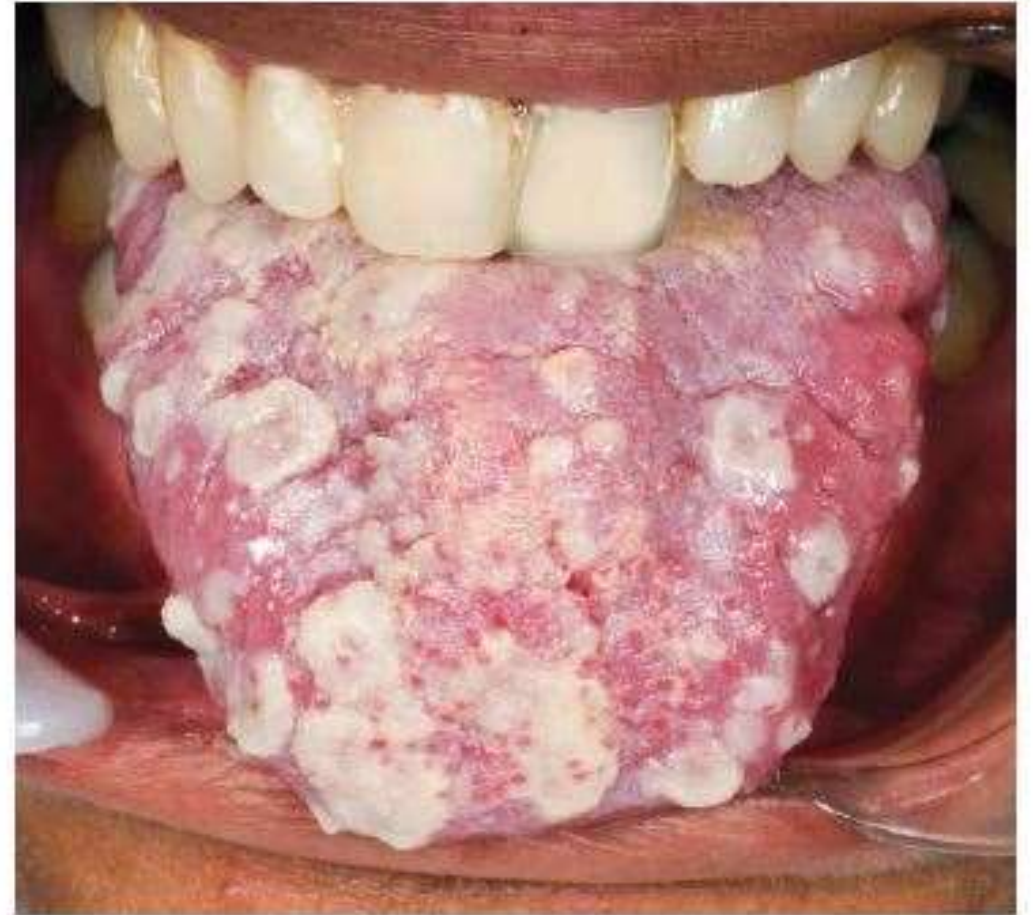


Figure 16.18 *Candida albicans* stomatitis.

Yeast infections treatment :

- The majority of superficial Candida infections can be treated using **topical** antifungals including clotrimazole, miconazole and nystatin.
- Many patients find **systemic** treatments more convenient such as fluconazole 150mg as a single dose or itraconazole 200mg twice for 1 day.

Deep Fungal Infections :

- Fungal infections of the deeper tissues are rare in healthy individuals and usually **only affect those who have underlying medical problems or are immunocompromised by illness or medication.**
- However, implantation of fungi into the skin through trauma may precipitate a chronic deep localized infection in otherwise healthy individuals.
- Fungal infections that involve the deeper skin tissues include histoplasmosis, cryptococcosis, sporotrichosis, and Fusarium spp.



Deep Fungal Infections :

- In immunocompromised, neutropenic and diabetic ketoacidosis patients, severe life-threatening infections may result from **mucormycosis** species such as **Rhizopus** (found in moldy bread).
- These fungi usually infect the facial sinus and may spread to the brain.
- Initially, symptoms can mimic sinusitis (nasal stuffiness) followed by facial swelling and black pus followed ultimately by necrotic tissue.



Systemic antifungal drugs:

- Severe or disseminated cutaneous Or systemic fungal infections may require prolonged treatment with a systemic antifungal drug.
- Oral preparations available include **terbinafine**, which can be used to treat widespread dermatophyte infections (scalp, nails, skin) and sporotrichosis.
- **Itraconazole** is effective against candidiasis, seborrheic dermatitis, Pityriasis versicolor, Aspergillosis, Histoplasmosis, and Blastomycosis.
- **Griseofulvin** is mainly used to treat tinea capitis but may also be used to treat tinea corporis.

Thank You